Presentation Recording

Mars Rover

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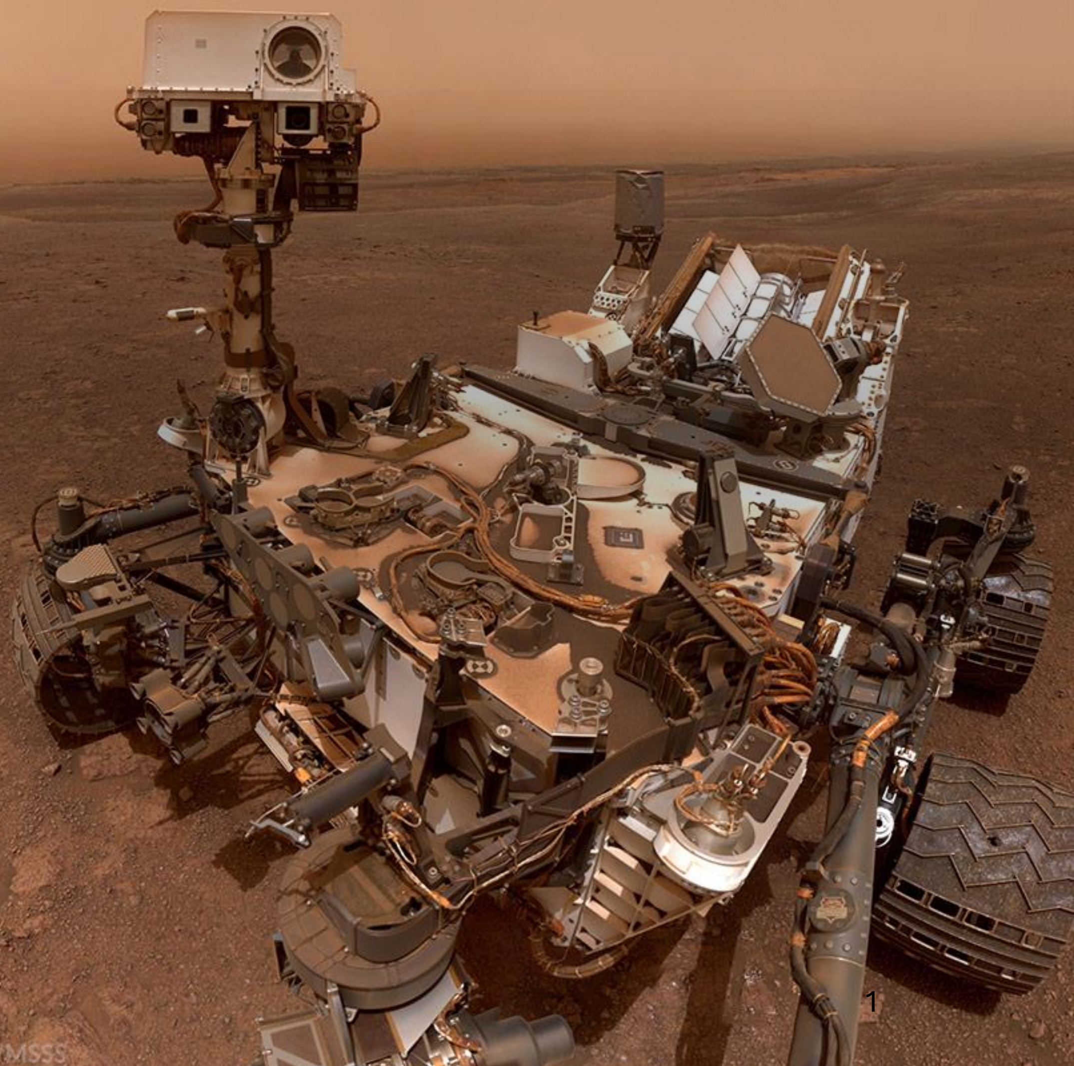


Curiosity rover on Mars - NASA/JPL-Caltech/MSSS

Overview

- Mission Overview
- Stakeholders
- Stakeholders Needs
- Requirements
- Functional Decomposition
- MBSE Artifacts
- Verification and Validation
- System Operations and Sustainment

Mission Overview

- *"To support sustained human presence on Mars, there is a need for a reliable logistics support systems capable of transporting supplies between landers, habitats, and surface assets. Human-driven logistics are impractical due to risk, time delay, and environmental hazards. This project presents an autonomous logistics rover, emphasizing mission operations, autonomy, safety, and sustainment."*

Stakeholders

This section will cover the following points:

- *Who are the Stake Holders*
- *What are Stake Holder Expectations*
- *Operational and Architectural Conflicts*

Stakeholders

- The primary organization leading this effort is NASA but with various subgroups as key stakeholders:
    - Mars Surface Operations: Responsible group for monitoring the system during its missions
    - Habitat and Science Users: Dependent on the system to provide logistical support
    - Systems Safety and Mission Assurance: Oversee that the system will not cause undue harm to any of the active stakeholders.
    - Maintainability and Sustainment: Ensures the system will serve as a successful logistical solution for many years to come

Stakeholder Expectations

Stakeholder	Expectations
NASA	• Reliable cargo transportation without crew involvement • Predictable operations under communications delay • Systems that continue to work after partial failures • Logistics capability that scales with surface operations
Mars Surface Operations Team	• Rover awareness of location and destination • Cargo delivered on time and intact • Confidence that the rover will not block crew operations • Clear operational modes and handoffs
Habitat and Science Users	• Dependable delivery schedule • Reliable payload protection from dust • Reliable payload protection from vibration • Adaptable mating interfaces
Systems Safety and Mission Assurance	• Predictable failure behaviors • Hazard mitigation for autonomous navigation • Confidence that the rover will not endanger the crew or habitat
Maintainability and Sustainment	• Increased reliability and minimal maintenance • Modular build to allow refurbishment • Autonomous failure detection • Autonomous self-recovery

Operational and Architectural Conflicts

- Operational
  - Users want adaptable physical integration, but MA wants a more conservative approach prioritizing safety and redundancy
  - Working autonomously with partial failures but should arrive on time and intact
  - Scalability of the system vs overall reliability
  - End user wants payload protected but Surface Ops want on payload delivered on time
- Architectural
  - Operates with minimal maintenance but needs modularity for future upgrades
  - Autonomy under communication delay vs predictability

Requirements

This section will cover the following points:

- Define system-level requirements for rover operations
- Derived requirements from subsystem faults and failure modes
- Requirements traceability to stakeholder needs and verification methods

Requirements

Requirement ID	Requirement Statement	Mission/System
AUTO-001	The fully functioning navigation system shall provide positional accuracy within 10 inches of truth.	System
AUTO-002	The navigation system shall provide directional accuracy within 1 degree of truth.	System
AUTO-003	The navigation system shall avoid commanded keep-out areas.	Mission
AUTO-004	The Rover shall use Image Processing and IMU telemetry for navigation.	System
AUTO-005	The Rover shall maintain a minimum separation distance of 10m from crew while navigating	System
AUTO-006	The Rover shall receive location data from crew	Mission
AUTO-008	The Rover shall automatically stop if a crew member enters the defined safety buffer	Mission
AUTO-009	The Rover shall enter a pre-determined mode if there is a failed self-test for navigation fidelity.	Mission
COM-001	The Rover shall accept commands for areas to avoid.	Mission
COM-002	The Rover shall have a physical switch for crew that will power off the Rover within 5 seconds.	System
COM-003	The Rover shall enter stand by mode if no communication has been received within 1 hr.	System
COM-004	The Rover shall receive a command to perform a self-test.	Mission
COM-005	The Rover shall receive a command to go to a target coordinate.	Mission
COM-006	The Rover shall receive a command to end the current task.	Mission
COM-007	The Rover shall send self-test results to operators each time they are performed.	Mission
COM-008	The Rover shall report historical location data since last transmission every 1hr.	System

Derived Requirements

- Subsystem level requirements are derived from identified failure modes to define how the individual subsystems detect, respond to, and recover from the faults during autonomous operations.

Requirement ID	Subsystem	Requirement Statement	Source Fault
NAV-SUB-001	Navigation (Camera)	The navigation subsystem shall detect camera obstruction by monitoring degradation in image quality or loss of visual features.	Camera Obstruction
NAV-SUB-002	Navigation (Camera)	The navigation subsystem shall command camera reorientation to avoid direct exposure to environmental debris when obstruction is detected.	Camera Obstruction
NAV-SUB-003	Navigation (Camera)	The navigation subsystem flags a navigation fault if visual obstruction persists beyond 5 mins.	Camera Obstruction
SW-SUB-001	Software	The software subsystem shall detect abnormal execution patterns such as Bootlooping using a hardware watchdog timer.	Bootlooping

Requirements Traceability Matrix

USER NEED ID	Requirement ID	Verification Method
UN-MSO-1: *We need the rover to know where it is and where it's going.*	AUTO-001	Instrumented Test
	AUTO-002:	Instrumented Test
	COM-005	Demonstration/Field Test
UN-MSO-2: *We need cargo delivered on time and intact*	AUTO-004	Instrumented Test
	COM-012	Demonstration/Field Test
	PHYS-001	Analysis
	PHYS-002	Demonstration/Field Test
	PHYS-007	Instrumented Test
	PHYS-012	Instrumented Test
	PHYS-013	Instrumented Test

- The Traceability Matrix serves as a tool to ensure that Requirements are tied to user needs and ensures there is a verification method pre-determined to validate the requirement has been met.

Functional Decomposition

This section will cover the following points:

- *Defines rover behavior through functional threads*
- *Mode-based operations*
- *External Interfaces and Internal Dependencies*

Functional Threads

The functional thread represent mission capabilities across subsystems.

- **Mission Tasking and Command Handling:** Processes commands and maintains mission state

Subsystems: TTC, CDH, Software, Fault Mgmt

- **Autonomous Navigation:** localization, path planning, hazard avoidance

Subsystems: GNC, Sensors, CDH, Software

- **Docking and Interface Operations:** Alignment and interaction with surface assets

Subsystems: GNC, Mobility, Mechanical, TTC, CDH and Cargo acquisition.

- **Cargo Handling and Transport:** Payload acquisition, protection, delivery

Subsystems: Payload, Mechanical, Mobility, Thermal control, and CDH

- **Communication & Data Handling:** Command exchange and telemetry

Subsystem: TTC, CDH, Software

- **Energy Management:** Battery monitoring and charging

Subsystem: Electrical, Power, CDH

- **System Health and Fault Response:** Fault detection, recovery, safe mode

Subsystems: CDH, Software, Mobility

- **Fault categories:** navigation, physical, software/communication

Mode-Based Operations

It defines the system behavior across mission phases and activates functional threads on system state.

Normal Mode:

- Idle/standby mode
- Mission Preparation mode
- Self-Navigation mode
- Docking and interface mode
- Cargo Transfer mode
- Charging Mode
- Diagnostic/maintenance mode

Fault Modes:

- Fault scan mode
- Software recovery mode
- Assisted motion mode
- Degraded mode
- Safe mode

Interface and Dependencies

External Interfaces

- Mission command & control (tasking and updates)
- Surface operations & crew systems (safety constraints)
- Surface assets (habitats, landers, charging stations)
- Communication network (telemetry & commands)
- Martian environment (terrain, dust, constraints)

Internal Dependencies

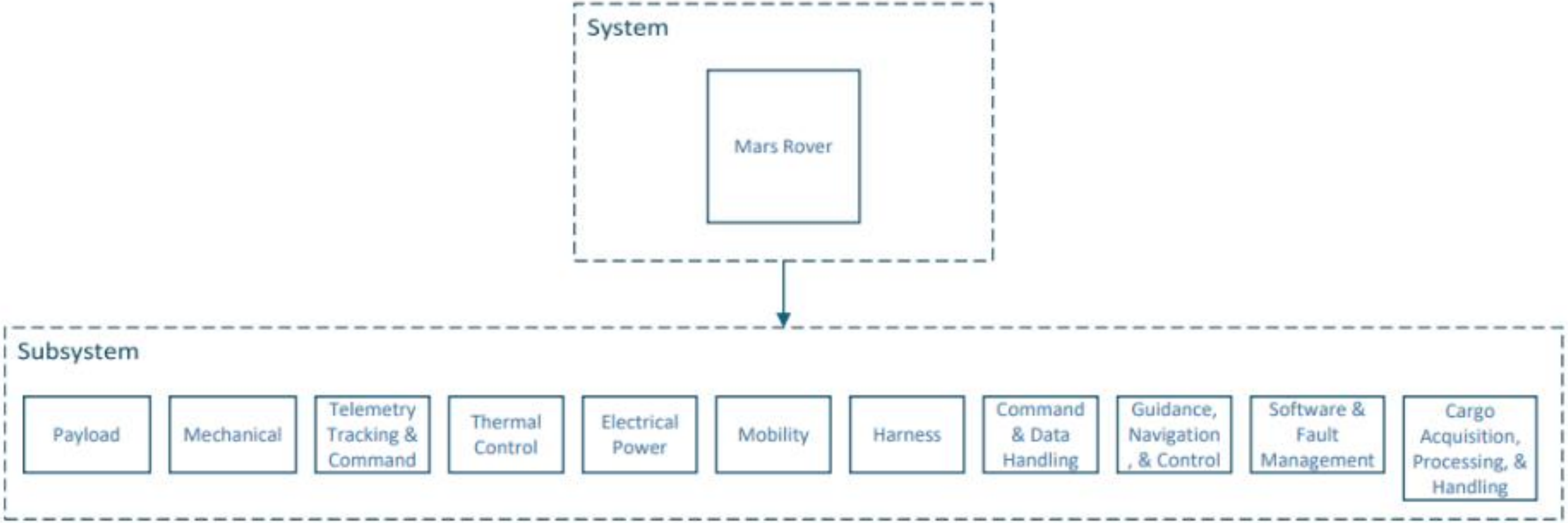
- Navigation → sensing, mobility, control
- Cargo → payload, mechanical, mobility
- Energy → power system & control logic
- Communication → TTC, CDH, software
- Fault handling → impacts multiple subsystems

MBSE Artifacts

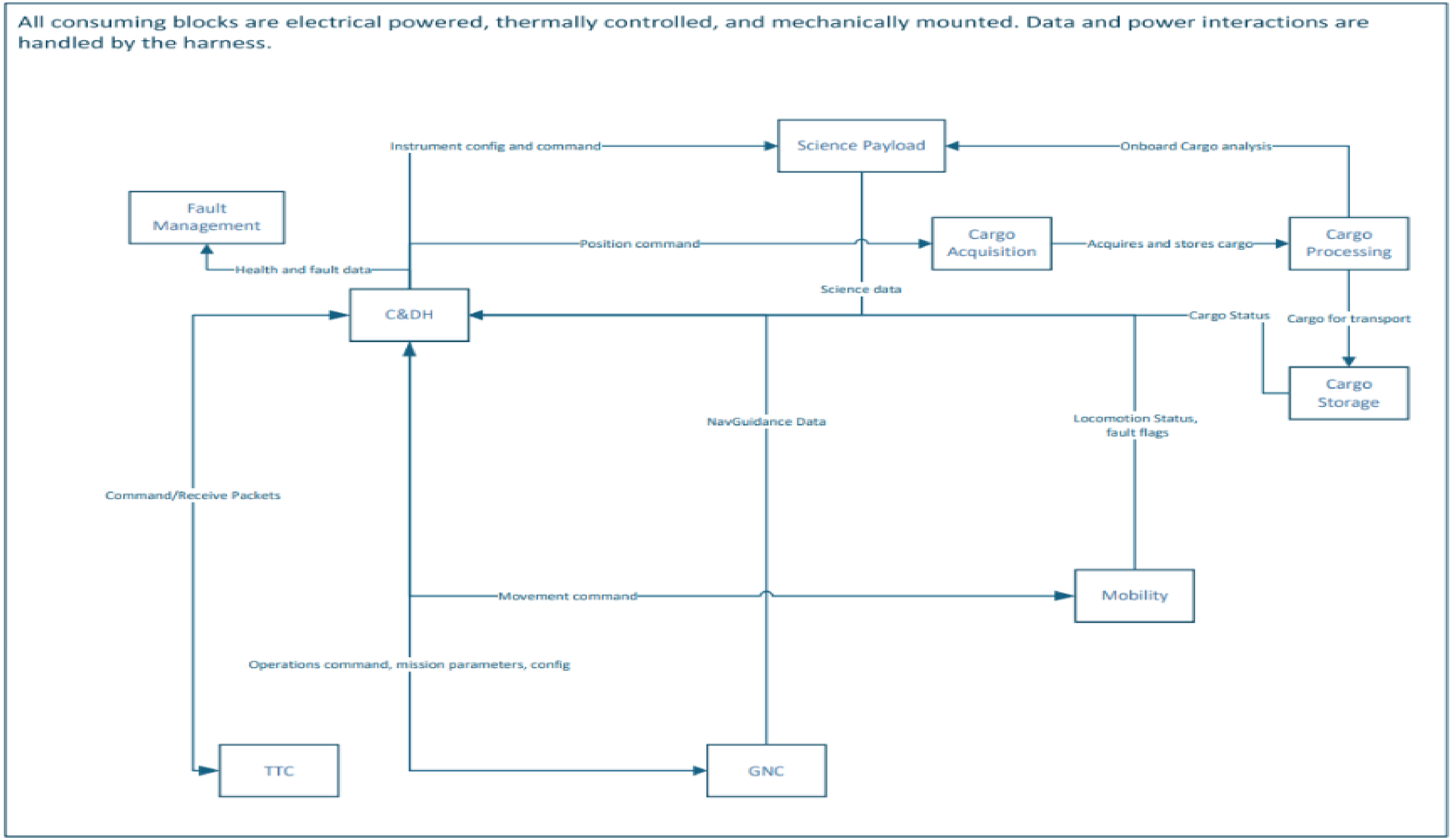
This section will cover the following points:

- *Represents system structure using visual models*
- *Illustrate interaction between subsystems*
- *Includes system context,* architecture *and functional relationships*.

Systems Architecture



Functional Interactions



Verification and Validation

This section will cover the following points:

- *Ensures system meets defined requirements*
- *Verifies performance using structures testing methods*
- *Validates system behavior under Mars mission conditions*

Verification Methods

The verification methods are used to evaluate system performance and confirm requirement satisfaction.

Methodes used are:

Instrumented Testing: Sensor based evaluation and fault response.
Subsystems: Navigation, Sensors, CDH, Software

Analysis: Simulation-based evaluation (thermal, structural)
Subsystem: Thermal, Electrical, Structural

Demonstration / Fielding Testing:
Realistic operations (nav, docking, cargo)
Subsystems: Mobility, Payload, Mechanical

Inspection: Hardware and Software compliance checks
Subsystems: Interfaces, components, code

Validation Under Mars Conditions

Validation ensures the rover performs efficiently under realistic Mars environmental and operational conditions.

- Uses emulator based and simulation testing
- **Autonomous Navigation:** validates operation under communication delay
- **Hazard detection:** validate navigation in dust and low visibility
- **Fault detection & recovery:** validates system response to failures
- **Cargo delivery:** validates performance under terrain constraints
- **Docking operations:** validates alignment and payload transfer
- **Thermal testing:** validates performance under extreme temperatures.

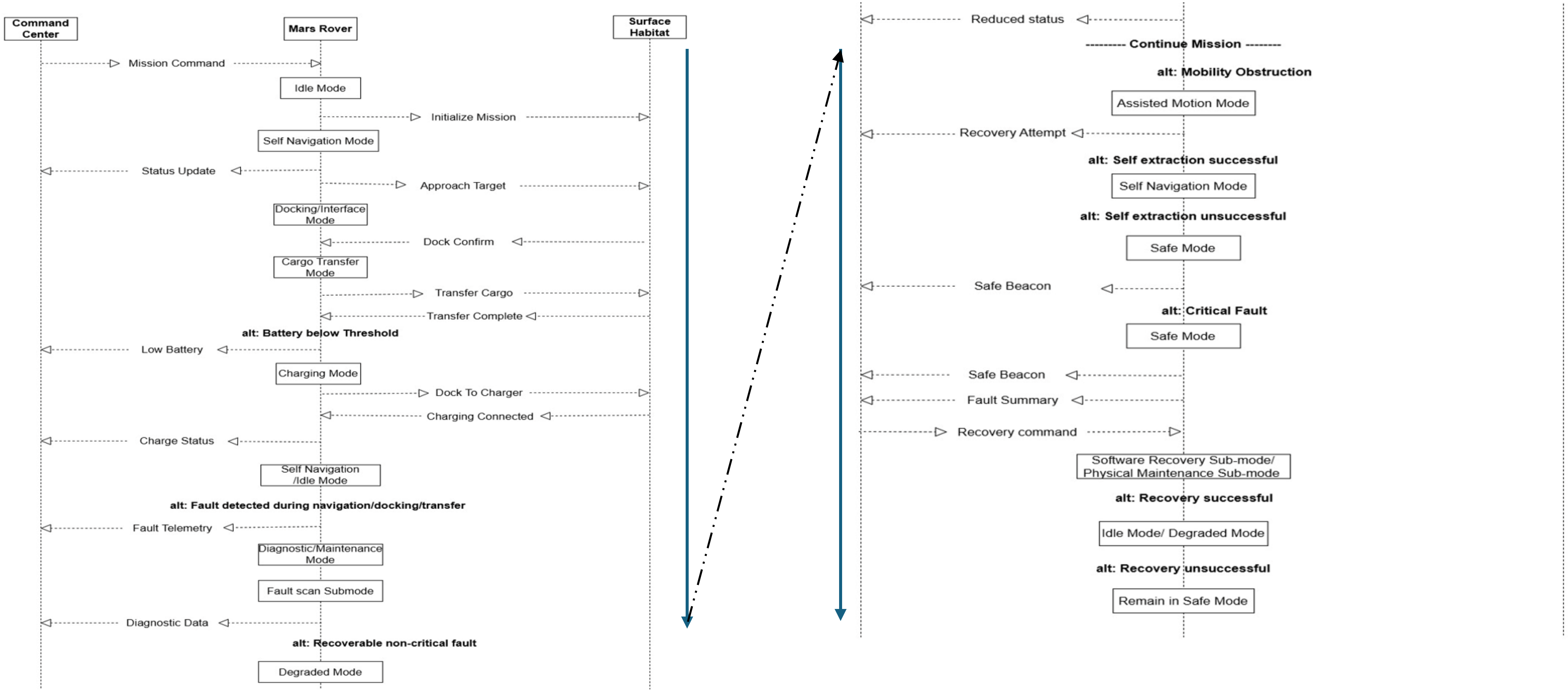
System Operations and Sustainment

This section will cover the following points:

- *Describes system operation across mission phases*
- *Autonomy and fault management*
- *Addresses long term reliability and sustainment*

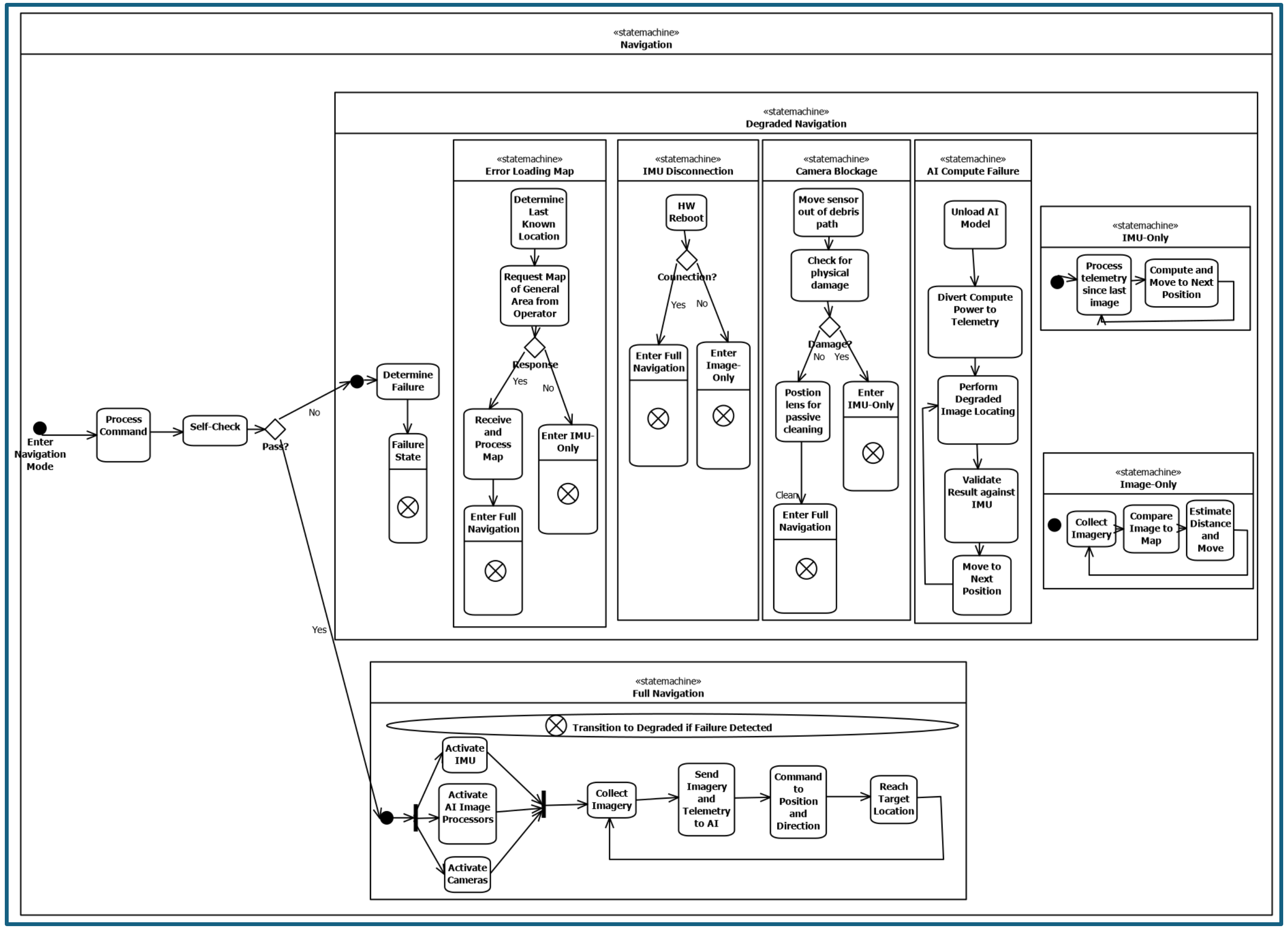
Mission Phase Mode Transitions

- Transitions between modes from Slide 14

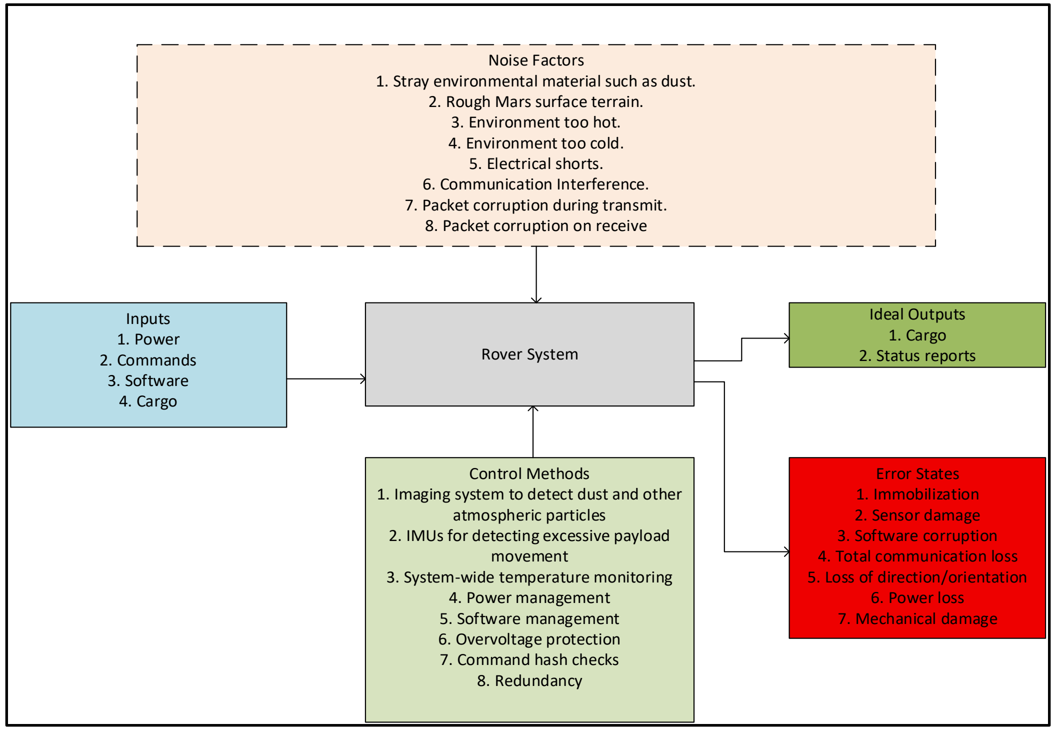


Reliability & Fault Management

*Capabilities and failure mode transitions – **Navigation***



System P-Diagram



Sustainment

- **Modular design** enabling future software and hardware upgrades
- **Customizable mission logic** ensures multi-mission integration potential

Potential changes over time